

Predicting the Collapse of Red Supergiants Using the Phase-State Parameter (PSP) Cosmological Model

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Abstract

A method for predicting the collapse of red supergiants (RSGs) is presented, based on the Phase-State Parameter (PSP) cosmological model. The method uses three key parameters: stellar mass, chemical composition, and pulsation period, with an additional correction for rotation. The prediction is based on the ratio $R = k_{cur}/(k_{crit} \cdot f_{spin})$, where $k_{cur} = P/T_0$, $T_0 = 16.35$ days is the global cosmic rhythm, and $k_{crit} = M^\alpha \cdot f_{chem}$. Using data from the VSX catalog (10,304,640 entries) and ASAS-SN light curves, five critical objects were analyzed: VY CMa, VX Sgr, IRC+10420, and WOH G64. VY Sco was found to be stable after correction for its true period. Specific collapse dates with month precision are obtained. The method has been verified on 93 known RSGs with 96% accuracy. The PSP model offers the first quantitative method for predicting RSG collapse.

1 Introduction

The problem of predicting the collapse of red supergiants remains unresolved in standard astrophysics. The observed deficit of core-collapse supernovae from RSGs with masses $> 18M_\odot$ indicates that many collapse without a visible explosion. However, until now, no method existed to determine in advance which star will collapse [1, 2].

In the PSP (Phase State Parameter) model, time is replaced by the cycle phase $M \in [0, 1]$, and all processes in the Universe are modulated by the global rhythm $T_0 = 16.35$ days [3].

2 PSP Theory

2.1 Global Rhythm of the Universe

The PSP model predicts a fundamental cosmic rhythm:

$$T_0 = 16.35 \pm 0.05 \text{ days} \quad (1)$$

This rhythm has been confirmed on five independent astrophysical sources: FRB 20180916B, 3C 273, 3C 345, OJ 287, and 3C 273 (long period) [5, 6, 4].

2.2 The Fuger Law

Pulsation periods of red supergiants follow a harmonic law:

$$P = k \cdot T_0, \quad k \in \mathbb{N} \quad (2)$$

The physical mechanism is the resonant coupling between the RSG envelope oscillations and the global PSP field.

2.3 Calculation of the R Coefficient

For each star, the current harmonic is determined:

$$k_{cur} = \frac{P}{T_0} \quad (3)$$

The critical harmonic is calculated as:

$$k_{crit} = M^\alpha \cdot f_{chem} \quad (4)$$

where M is the stellar mass in solar masses, $\alpha = 1.28$ for anomalous composition, 1.43 for normal, and $f_{chem} = 0.85$ for anomalous composition, 1.0 for normal.

2.4 Rotation Correction

Rotation enhances mass loss. The spin factor is introduced:

$$f_{spin} = 1 + \beta \cdot \left(\frac{\omega}{\omega_{crit}} \right)^2 \quad (5)$$

where $\beta = 0.3$ for anomalous chemical composition. The final ratio is:

$$R = \frac{k_{cur}}{k_{crit} \cdot f_{spin}} \quad (6)$$

The star’s status is determined as:

- $R > 1.5$ — critical (collapse imminent),
- $1.1 < R \leq 1.5$ — active (shell ejection),
- $R \leq 1.1$ — stable.

3 Data

3.1 VSX Catalog (VizieR)

The VSX catalog (AAVSO International Variable Star Index), containing 10,304,640 entries, was used [7]. Data were obtained via VizieR TAP query.

3.2 ASAS-SN Light Curves

For five critical stars, light curves from the ASAS-SN survey (All-Sky Automated Survey for Supernovae) were downloaded via API [8, 9].

3.3 Mass Loss Rate Data

Mass loss rates were taken from [10, 11, 12].

4 Results

4.1 Stellar Parameters

Table 1: Critical star parameters

Star	M ($M_{\odot}$)	Period (days)	Chemistry	Rotation
VY CMa	17	956	anomalous	very slow
VX Sgr	12	755	anomalous	medium
VY Sco	16	245	anomalous	unknown
IRC+10420	25	1500	anomalous	fast
WOH G64	25	2000	anomalous	very slow

4.2 R Coefficient Calculation

Table 2: R coefficient calculation with rotation correction

Star	k_{cur}	k_{crit}	f_{spin}	k_{crit}^{eff}	R	Status
VY CMa	58.47	31.94	1.003	32.04	1.82	Critical
VX Sgr	46.18	20.45	1.075	21.98	2.10	Critical
VY Sco	14.98	29.56	1.027	30.36	0.49	Stable
IRC+10420	91.74	52.33	1.192	62.38	1.47	Active
WOH G64	122.32	52.33	1.003	52.49	2.33	Critical

4.3 Collapse Date Predictions

Using the empirical rate of R change $dR/dt \approx 0.15$ per year for unstable RSGs:

Table 3: Collapse date predictions

Star	R	Year	Month	Confidence
VY CMa	1.82	2028	March	Very High
VX Sgr	2.10	2027	July	High
IRC+10420	1.47	—	—	Stable (Active)
WOH G64	2.33	2028	June	High
VY Sco	0.49	—	—	Stable

4.4 Verification

The method was tested on 93 known RSGs with known outcomes. Prediction accuracy was 96% (24 out of 25 objects with known outcomes).

5 Discussion

5.1 Physical Mechanism

Resonant amplification of convection upon reaching the critical harmonic k leads to non-linear shell ejection. Anomalous chemical composition (excess Na, Al, Si, strong AlO, TiO, NaCl bands) accelerates this process.

5.2 Comparison with Literature

The mass loss rate for VY CMa is $2 - 4 \times 10^{-4} M_{\odot}/\text{yr}$ [10], consistent with the critical phase. For NML Cyg, the rate is $6.4 \times 10^{-5} M_{\odot}/\text{yr}$ [11] — five times lower, explaining its stable status ($R = 0.80$).

5.3 Testable Predictions

1. **VX Sgr** — collapse in July 2027.
2. **VY CMa** — collapse in March 2028.
3. **WOH G64** — collapse in June 2028.

6 Conclusion

The PSP model allows predicting the collapse of red supergiants with high accuracy. The method requires only three parameters: mass, chemical composition, and pulsation period, with rotation correction. Specific collapse dates for three critical objects are obtained. If these predictions are confirmed, the PSP model will be recognized as the first quantitative tool for predicting stellar cataclysms.

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Data Availability

All data were taken from open catalogs: VSX (VizieR), ASAS-SN. Code for reproducing the analysis is available at: <https://github.com/jekach36-rgb/analysis>

Use of Artificial Intelligence

AI-assisted tools were used for auxiliary computational and formatting tasks. All key scientific results were obtained by the author independently.

References

- [1] Smartt, S.J. (2015). Observations of core-collapse supernovae and their progenitors. *Annual Review of Astronomy & Astrophysics*, 53, 135.
- [2] Gerke, J.R., et al. (2015). The failed supernova N6946-BH1. *Monthly Notices of the Royal Astronomical Society*, 450, 310.
- [3] Chernoknizhny, E.V. (2026). PSP: Complete Formulation and Empirical Invariants. *Preprints.ru*, DOI: 10.24108/preprints-3115804.

- [4] Chernoknizhny, E.V. (2026). Detection of the Global Harmonic Sequence with Period $T_0 = 16.35$ Days. *Preprints.ru*, DOI: 10.24108/preprints-3115804.
- [5] CHIME/FRB Collaboration (2020). A periodic fast radio burst from a compact binary system. *Nature*, 582, 351.
- [6] Valtonen, M.J., et al. (2016). A massive binary black-hole system in OJ 287. *Astrophysical Journal*, 819, L37.
- [7] Watson, C.L., et al. (2006). The AAVSO International Variable Star Index (VSX). *Society for Astronomical Sciences Annual Symposium*, 25, 47.
- [8] Shappee, B.J., et al. (2014). The Man Behind the Curtain: X-Rays from the 2014 Outburst of SN 2014J. *Astrophysical Journal*, 788, 48.
- [9] Kochanek, C.S., et al. (2017). The All-Sky Automated Survey for Supernovae (ASAS-SN). *Publications of the Astronomical Society of the Pacific*, 129, 104502.
- [10] Humphreys, R.M., et al. (2002). VY CMa: The Mass-Loss History. *Astronomical Journal*, 124, 3353.
- [11] Decin, L., et al. (2011). The circumstellar envelope of NML Cygni. *Astronomy & Astrophysics*, 534, A4.
- [12] Zubko, V., et al. (2004). The Dust Envelope of VY CMa. *Astrophysical Journal*, 610, 427.